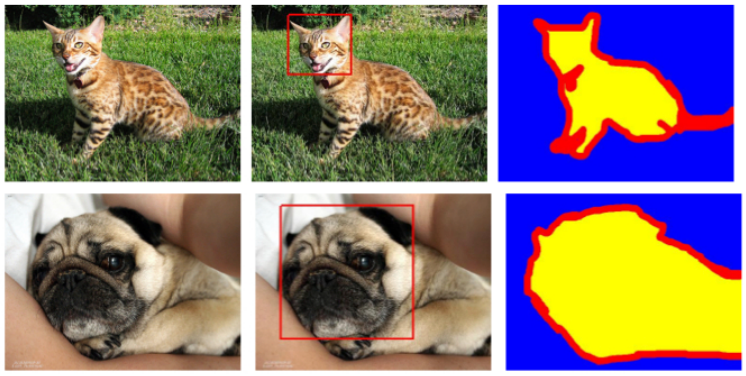
Assignment 1 – Image Segmentation Implementation

11376/8890 – Computer Vision and Image Analysis (CVIA)

Context

This assignment focuses on image segmentation, a core task in computer vision that aims to partition an image into meaningful regions at the pixel level. You are required to implement and compare different segmentation methods, gaining insight into how these vision techniques differ in behaviour and performance.

Dataset



We will use the **Oxford-IIIT Pet** dataset. It is a widely used benchmark dataset for image classification and segmentation tasks. It contains approximately 7,000 high-resolution images of cats and dogs across 37 distinct breeds. Each image is accompanied by a pixel-level annotation in the form of a tri-map, which labels regions as foreground (the pet), background, and boundary. The dataset features significant variation in pose, scale, lighting, and background complexity, making it suitable for evaluating the robustness of segmentation algorithms. Its relatively moderate size and high-quality annotations make it particularly well-suited for experimentation with both classical and deep learning-based segmentation methods.

- ~7,000 images of cats and dogs
- Pixel-wise segmentation masks are provided (foreground/background) and boundary.
- Available on *Canvas -> Modules -> Assignment 1 Information*

For our assignment, we will use a small subset of Oxford-IIIT Pet, named **Oxford-IIIT Pet Subset**, which contains around 100 images extracted from the original dataset.

- ~100 images of 5 cats and dogs breeds (~20 images for each class)
- Available on *Canvas -> Modules -> Assignment 1 Information*
- The “image” folder contains all the images.
- The “annotation” folder contains the associated ground truth (segmentation map) for each image. Pixel Annotations: 1 - Foreground; 2 - Background; 3 - Not classified.
- You can treat the task as binary segmentation using the masks provided in the annotation folder.
- Use the “README” file for information of the dataset.

Task 1 – Edge-Based Segmentation (25 marks)

- Select one edge-based segmentation algorithm
- Describe the fundamental principles and core idea of the selected method (5 marks)
- Implement the selected method (10 marks)
    - Examples: Canny + contour extraction, Sobel/Laplacian + thresholding
    - Convert to segmentation map
- Present results (10 marks)
    - Quantitative measures (IoU/Dice)
    - Qualitative results (visual examples)

Task 2 – Region-Based Segmentation (25 marks)

- Select one region-based segmentation algorithm
- Describe the fundamental principles and core idea of the selected method (5 marks)
- Implement the selected method (10 marks)
    - Examples: Region Gowing, Watershed
    - Define similarity criteria and justify parameters
- Present results (10 marks)
    - Quantitative measures (IoU/Dice)
    - Qualitative results (visual examples)

Task 3 – Clustering-Based Segmentation (30 marks)

- Select one clustering -based segmentation algorithm
- Describe the fundamental principles and core idea of the selected method (5 marks)
- Implement the selected method (15 marks)
    - Examples: K-means, Mean Shift, GMM
    - Experiment with different features (RGB/HSV/spatial features)
    - Justify your selection
- Present results (10 marks)
    - Quantitative measures (IoU/Dice)
    - Qualitative results (visual examples)
    - Comparison of at least two different features

Task 4 – Comparison and your insights (10 marks)

- Compare these methods and their performance
- Discuss your observations
- Summaries your insights

Good Programming (5 marks)

Demonstrated the use of good programming principles such as clear program structures, good use of comments, use of functions (where appropriate) etc.

Bonus (Up to 5 marks)

Implement an extension such as combining methods, pre-processing or post-processing improvements. You need to clearly state which part is the extension in your report.

Evaluation Metrics

Each pixel is classified as: foreground (pet) or background, so the evaluation becomes a pixel-wise comparison:

	Ground Truth: Foreground	**Ground Truth: Background**
Predicted Foreground	True Positive (TP)	False Positive (FP)
Predicted Background	False Negative (FN)	True Negative (TN)

We can then define the following two evaluation metrics:

(1) Intersection over Union (IoU)
IoU measures how well the predicted segmentation overlaps with the ground truth. It is defined as the ratio between the area of overlap and the area of union between the predicted mask and the ground truth mask:

$$IoU = \frac{TP}{TP + FP + FN}$$

IoU provides a value between 0 and 1, where 1 indicates perfect segmentation and 0 indicates no overlap. A higher IoU indicates better segmentation performance.

(2) Dice Coefficient (F1 Score)
It is similar to IoU, but slightly more forgiving. The Dice Coefficient measures the similarity between the predicted segmentation and the ground truth by considering their overlap. It is calculated as twice the area of overlap divided by the total number of pixels in both masks:

$$Dice = \frac{2TP}{2TP + FP + FN}$$

A higher Dice score indicates better performance, with 1 indicating a perfect match.

In your experiments, present quantitative results (e.g., average IoU or/and Dice score), and include qualitative visual examples to illustrate the segmentation performance.

Submission Requirements

- **A PDF report** – single line spacing, font size 10, maximum 5 pages. It should include:
  - Method Description: Explain each selected algorithm and their parameters.
  - Results and Comparison: Quantitative metrics and visual comparisons.
  - Discussion: Strengths, weaknesses, potential improvements.

- **Python code** – everything in a single Zip file, well-documented with instructions to run the code (if necessary)

Notes

- Tune parameters carefully using a validation set.
- Recommended libraries: OpenCV, scikit-image, NumPy, etc.